

STATEMENT OF WORK
For
Verdi V6 Raman 532 nm laser
2/13/2025

1.0 INTRODUCTION

Naval Surface Warfare Center Indian Head Division (NSWC IHD) requires the acquisition of a Verdi V6 532 nm (green) laser. This instrument will be used for the Raman spectroscopy analysis of microscopic samples, both at ambient conditions and at high pressure-high temperature conditions. The vendor shall provide the laser with a stable wavelength and adjustable power, permanently fixed optical elements and aluminum-free diode bars. The instrument shall deliver 6W of power and be suitable for upgrade to higher power for future experiments.

2.0 REQUIREMENTS

Laser

- a. Must produce 6W of 532 nm laser light
- b. Single frequency (<5 MHz over 50 msec)
- c. Noise <0.02% rms from 10 Hz to 1 GHz
- d. Diffraction-limited beam with $M^2 < 1.1$
- e. Divergence < 0.5 mrad
- f. Power stability $\pm 1\%$
- g. Laser head must be sealed
- h. Must be able to minimize diode drive current
- i. All optical elements must be permanently fixed (solder bonded) and aligned
- j. Must have aluminum-free diode bars

Chiller and power supply

- k. Size should not exceed 1 ft³ in volume
- l. Power supply must allow the adjustment of output power
- m. Size of the power supply should not exceed 17.2 x 17.8 x 7 inches

The Contractor shall make delivery 16 weeks after awarding the contract.

3.0 WARRANTY

Upon government ownership of the instrument, the standard warranty will begin. The Contractor shall provide warranty to include replacement or repair of any hardware except for consumable items.

4.0 ACCESS ON SITE

The Contractor delivery personnel are required to be US Citizen and provide documentation on driver for access and delivery destination point.

5.0 DELIVERY

The Contractor shall drop-shipped directly to B3146 via NSWC IHD riggers. The customer will perform all installation.

NSWC IHD

RECEIVING

ATTN.: Don Bragunier/ 240-695-3124

4522 McMahon Road, B116

Indian Head MD 20640-5125

6.0 TECHNICAL POINT OF CONTACT

The Contractor shall communicate with the TPOC during development, shipment and delivery.

NSWC IHD

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NSWC IHD

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